

Every Coin Counts

Counting coins on a textured pad with classical image processing

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90.29%

count accuracy
33 coins off out of 340

1.27

mean abs. error
coins per photo

11/26

exact photos
19/26 within ± 1 coin

9/12

buttons excluded
by holes or colour

The problem

People count coins at a glance; a computer sees only pixels. Our phone photos add three traps: a white pad full of folds and seams, dark or motion-blurred shots, and buttons the size of a coin mixed in.

Question Can fixed classical rules, with no training, count only the coins under these conditions?

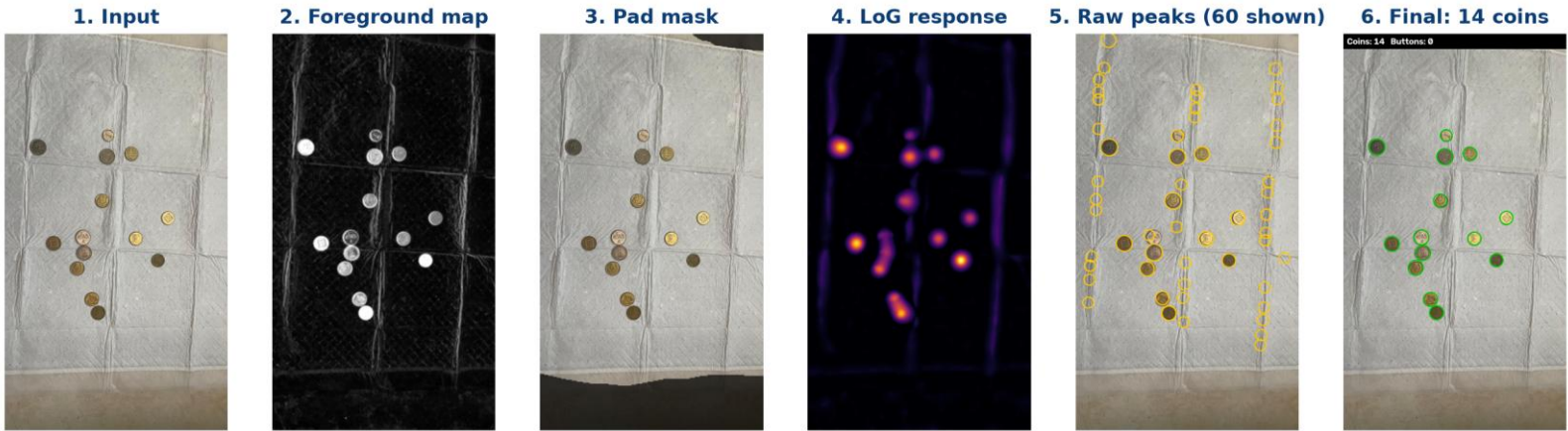
Data 26 photos · 340 coins · 12 buttons, hand-counted and split into clean, noisy (dark / blurred) and damaged (piles, cut coins, buttons). A synthetic image with 6 discs checked the code: 6 of 6 found.

The pipeline



- **Background:** a huge median filter models pad and light; each pixel gets a Lab “difference from background” score.
- **Blobs:** multi-scale Laplacian of Gaussian finds round, coin-sized peaks.
- **Folds:** a Hessian test rejects ridges; a grayscale opening removes thin seams.
- **Touching coins:** Hough circles split what LoG merges.
- **Buttons:** black-hat reveals sewing holes; blue or pink objects are never coins.

From photo to count



Input → foreground map → pad mask → LoG response → raw peaks (mostly folds) → 14 coins after verification.

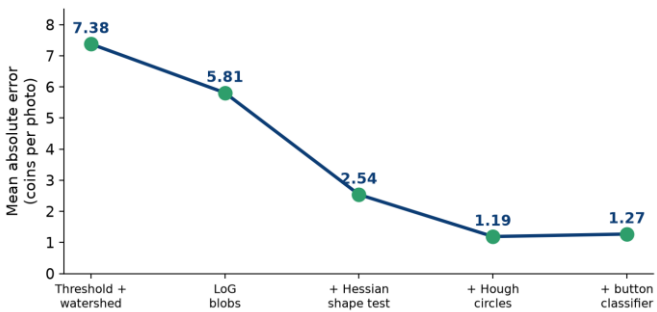
Results



Exact counts on clean and dark photos.

Clean photos: MAE 0.13, all within one coin. About 0.7 s per 1600 × 900 photo.

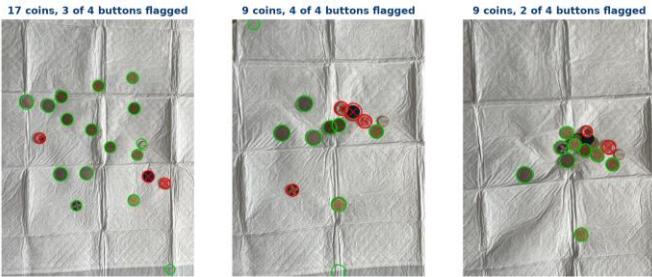
What made it work



Error after each development step.

Pad folds fooled every early version. The Hessian test, round versus ridge, cut the error by more than half.

Coins vs. buttons



Red X = button, not counted.

Blue, navy and pink buttons fail the colour test; for the rest, black-hat reveals 2–4 sewing holes around the centre.

Conclusion

Classical tools alone reached 90.29% count accuracy and excluded 9 of 12 buttons. Shape (Hessian) beat brightness for ignoring folds, and LoG + Hough together handled blurry and touching coins.

Limitations

Strong motion blur splits a coin into several blobs (22 vs 14). Dense piles hide coins, coins cut by the frame are half-circles, and a grey button without visible holes looks like a coin.

Next steps

Merge detections along the blur direction, fit partial circles at the image border, use rim colour for grey buttons, and mark coin positions to separate false positives from misses.